Lecture 7: Delta Method and Hypothesis Testing

IEOR E4709: Statistic Analysis and Time Series
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1 The Delta Method

This part is not covered in John Rice's textbook. You can consult Chapter 3 of the book "Asymptotic Statistics' by A. W. van der Vaart, from the Vrije Universiteit, Amsterdam. If you use your Columbia account, this book should be available at `https://www.cambridge.org/core/books/asymptotic-statistics/A3C7DAD3F7E66A1FA60E9C8FE132EE1D`.

Motivation: Suppose we observe $X_1, \ldots, , X_n$ independent Bernoulli(p) random variables. We are interested in estimating the odds, i.e., in the ratio $\frac{p}{1-p}$. For example, if the success of a medical treatment has probability $p = 2/3$, the odds of getting better is $2:1$. Furthermore, if there is another treatment with success probability r, then we might be interested in the odds ratio $\frac{p}{1-p}/\frac{r}{1-r}$, which gives the relative odds of one treatment over another.

Question: Suppose we use $\frac{\hat{p}_{MLE}}{1-\hat{p}_{MLE}}$ to estimate the odd $\frac{p}{1-p}$. What is the variance of $\frac{\hat{p}_{MLE}}{1-\hat{p}_{MLE}}$? Can we approximate the distribution of this sampling estimator?

Let $T_1, \ldots, , T_d$ be unbiased estimators (each T_i can be a function of random variables) of the parameters $\theta_1, \ldots, \theta_d$, respectively. Define $T = (T_1, \ldots, T_d)$ and $\theta = (\theta_1, \ldots, \theta_d)$. Consider a differentiable function $g(\cdot)$, and define the partial derivatives as

$$g_i'(\theta) = \left.\frac{\partial g(t)}{\partial t_i}\right|_{t_1=\theta_1,\ldots,t_d=\theta_d}.$$

The first-order Taylor expansion of g around θ is

$$g(t) = g(\theta) + \sum_{i=1}^{d} g_i'(\theta)(t_i - \theta_i) + Remainder$$

Let us construct a statistical approximation for θ using the estimator vector T. Then

$$g(T) \approx g(\theta) + \sum_{i=1}^{d} g_i'(\theta)(T_i - \theta_i)$$

Taking expectations on both sides, and using the equations above, we get

$$\begin{aligned}
\mathbb{E}\left[g(T)\right] &\approx g(\theta) + \sum_{i=1}^{d} g_i'(\theta)\mathbb{E}\left[T_i - \theta_i\right] \\
&= g(\theta) \\
Var\left[g(T)\right] &\approx \mathbb{E}\left[(g(T) - g(\theta))^2\right] \\
&\approx \mathbb{E}\left[\left(\sum_{i=1}^{d} g_i'(\theta)(T_i - \theta_i)\right)^2\right] \\
&= \sum_{i=1}^{d} g_i'(\theta)^2 Var(T_i) + 2\sum_{i>j} g_i'(\theta) g_j'(\theta) Cov(T_i, T_j),
\end{aligned}$$

where the last equality follows by expanding the square. Back to our example of the odds ratio estimator. We take $g(p) = \frac{p}{1-p}$ so that $g'(p) = \frac{1}{(1-p)^2}$. In this case $d = 1$, thus there is only one derivative. The MLE of p is $T_1 = \hat{p}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} X_i$. Thus

$$Var\left[g(T_1)\right] \approx (g'(p))^2 Var[\hat{p}_{MLE}] = \left(\frac{1}{(1-p)^2}\right)^2 \frac{p(1-p)}{n} = \frac{p}{n(1-p)^3} = \frac{(g'(p))^2}{I(p)},$$

where we recall from lecture 5 that $\frac{1}{p(1-p)}$ is the Fisher information of the parameter p for a single Bernoulli random variable, so that $I(p) = \frac{n}{p(1-p)}$ is the Fisher information of the entire data sample consisting of n independent samples. This gives an approximation for the variance of our estimator.

We can formulate a general result. Recall that the MLE estimator is asymptotically efficient and normal, i.e.

$$I(\theta)^{1/2}\left(\hat{\theta}_{MLE} - \theta\right) \xrightarrow{D} N(0, I),$$

where I is the identity matrix. $I(\theta)$ is the Fisher information associated with the whole sample. Then, we have that

Proposition 1.1.

$$g(\hat{\theta}_{MLE}) - g(\theta) \sim^D N\left(0, \nabla_\theta g(\theta)^\top I^{-1}(\theta)\nabla_\theta g(\theta)\right),$$

where $\nabla_\theta g(\theta) = (\frac{\partial g}{\partial \theta_1}, \frac{\partial g}{\partial \theta_2}, \ldots, \frac{\partial g}{\partial \theta_d})^\top$ *is the vector of partial derivatives. Above, the symbol* $\sim^D$ *means approximately equal in distribution as the number of observations grows large.*

2 Some Remarks on the Estimators

The material in this section will not be discussed in class, but you are strongly encouraged to read it.

A popular model for describing the dynamics of a stock price is the geometric Brownian motion

$$dS(t) = \mu S(t)dt + \sigma S(t)dW(t),$$

where $W(t)$ is a Brownian motion. Suppose that we can observe the stock price continuously. This is, of course, a stylized setup but frequently used in finance in the context of derivative pricing.

Also, the continuous assumption is a good approximation if we are dealing with high-frequency data. We would like to get the MLE estimates of the model parameters μ and σ. Using Ito's lemma, we can show that

$$d\log S(t) = \nu dt + \sigma dW(t),$$

where $\nu = \mu - \frac{\sigma^2}{2}$. Based on this relation, it is enough to find the MLE of ν and σ^2. Suppose we observe the stock price in equally spaced time periods $t_0 < t_1 < \cdots < t_n$, where $\delta := t_i - t_{i-1}$. Then $X_i := \log S(t_i) - \log(S_{t_{i-1}})$, $i = 1, \ldots, n$, are i.i.d. Gaussian samples of the Gaussian distribution with mean $\nu\delta$ and variance $\sigma^2\delta$. Recall the log-likelihood function of a Gaussian. In our case, it is given by

$$\log L(X\big|\nu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\delta\sigma^2) - \frac{1}{2\sigma^2\delta}\sum_{i=1}^{n}(X_i - \nu\delta)^2,$$

where $X = (X_1, X_2, \ldots, X_n)$. Using the first order condition, that is differentiating and setting the derivative equals to zero, yields the MLE estimates

$$\widehat{\nu}_{MLE} = \frac{1}{\delta}\bar{X}, \qquad \widehat{\sigma^2}_{MLE} = \frac{1}{n\delta}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

We first fix $\delta > 0$ (this can be one day, for instance, i.e. we only get end-of-day price observations) and send $n \to \infty$. We know that, by the properties of the MLE estimator, that the MLE is a consistent estimator of ν and σ.

In recent years, financial data such as stock prices are available at a very high frequency, i.e., δ can be infinitesimally small (on the order of microseconds). Fix $T = \delta n = t_n - t_0$ and let δ go to zero. Notice that

$$\begin{aligned}\hat{\nu}_{MLE} &= \frac{1}{\delta n}\sum_{i=1}^{n} X_i = \frac{1}{\delta n}\sum_{i=1}^{n}\left(\log S(t_i) - \log(S_{t_{i-1}})\right) \\ &= \frac{1}{\delta n}\left(\log S_{t_n} - \log S_{t_0}\right) = \frac{1}{T}\left(\log S_{t_n} - \log S_{t_0}\right)\end{aligned}$$

Hence, $\hat{\nu}$ does not converge in probability to ν as $\delta \to 0$.

Consider, instead, the estimator for the squared-volatility

$$\begin{aligned}\widehat{\sigma^2}_{MLE} &= \frac{1}{\delta n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 = \frac{1}{n\delta}\left[\sum_{i=1}^{n} X_i^2 - n(\bar{X})^2\right] \\ &= \frac{1}{\delta n}\left(\sum_{i=1}^{n}\left(\log S(t_i) - \log(S_{t_{i-1}})\right)^2\right) - \delta\hat{\nu}^2\end{aligned}$$

It is well known (from stochastic calculus) that, as $\delta \to 0$, $\left(\sum_{i=1}^{n}\left(\log S(t_i) - \log(S_{t_{i-1}})\right)^2\right)$ is the quadratic variation of $\log S$ in $[t_0, t_n]$, which is almost surely $\sigma^2(t_n - t_0) = \sigma^2 T$.[1] Hence, $\widehat{\sigma^2}_{MLE}$ converges almost surely to σ^2, as $\delta \to 0$ (notice that $\delta\hat{\nu}^2_{MLE}$ converge almost surely to zero as $\delta \to 0$ because the random variable $\hat{\nu}^2_{MLE}$ is almost surely bounded). This means that high-frequency data improve the accuracy of the estimate of the volatility σ, but not the accuracy of the estimate of the growth rate of the stock μ. Indeed, it is difficult to estimate expected stock returns.

[1] The proof that the quadratic variation of a Brownian motion $W(t)$ is almost surely t can be found, for instance, in Shreve's textbook "Stochastic Calculus for Finance II", see Theorem 3.4.3 therein.